Cauchy-Schwarz Masterclass

Problems

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1. Starting with Cauchy 1.1. The 1-Trick and the Splitting Trick

By the Cauchy-Schwarz inequality, with $b_1 = \dots = b_n = 1$, we get

$$a_1 + a_2 + \dots + a_n \le \sqrt{n}(a_1^2 + a_2^2 + \dots + a_n^2)^{\frac{1}{2}}.$$

For the second inequality, consider $x_k = a_k^{\frac{1}{3}}$ and $y_k = a_k^{\frac{2}{3}}$. Then, by Cauchy-Schwarz we have

$$\begin{aligned}\sum_{k=1}^n x_k y_k &\le \left(\sum_{k=1}^n x_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^n y_k^2\right)^{\frac{1}{2}} \\ \implies \sum_{k=1}^n a_k &\le \left(\sum_{k=1}^n a_k^{\frac{2}{3}}\right)^{\frac{1}{2}} \left(\sum_{k=1}^n a_k^{\frac{4}{3}}\right)^{\frac{1}{2}}\end{aligned}$$

1.2. Product of Averages and Averages of Products

By the Cauchy-Schwarz inequality and using the fact that $1 \le a_k b_k$ implies $1 \le \sqrt{a_k b_k}$, we have

$$\left(\sum_{k=1}^n p_k a_k\right)\left(\sum_{k=1}^n p_k b_k\right) \ge \left(\sum_{k=1}^n p_k a_k^{\frac{1}{2}} b_k^{\frac{1}{2}}\right)^2 \ge \left(\sum_{k=1}^n p_k\right)^2 = 1.$$

1.3. Why Not Three or More?

By applying the Cauchy-Schwarz inequality twice,

$$\begin{aligned}\sum_{k=1}^n a_k b_k c_k &\le \left(\sum_{k=1}^n a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^n b_k^2 c_k^2\right)^{\frac{1}{2}} \\ &\le \left(\sum_{k=1}^n a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^n b_k^4\right)^{\frac{1}{4}} \left(\sum_{k=1}^n c_k^4\right)^{\frac{1}{4}}.\end{aligned}$$

Thus,

$$\left(\sum_{k=1}^n a_k b_k c_k\right)^4 \le \left(\sum_{k=1}^n a_k^2\right)^2 \sum_{k=1}^n b_k^4 \sum_{k=1}^n c_k^4.$$

For the next inequality, we start with the observation that for any $k \in [n]$,

$$c_k \le \left(\sum_{k=1}^n c_k^2\right)^{\frac{1}{2}}.$$

Using this and the Cauchy-Schwarz inequality, we have

$$\begin{aligned}\sum_{k=1}^n a_k b_k c_k &\le \sum_{k=1}^n a_k b_k \left(\sum_{k=1}^n c_k^2\right)^{\frac{1}{2}} \\ &\le \left(\sum_{k=1}^n a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^n b_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^n c_k^2\right)^{\frac{1}{2}}.\end{aligned}$$

Thus,

$$\left(\sum_{k=1}^{n} a_k b_k c_k\right)^2 \leq \sum_{k=1}^{n} a_k^2 \sum_{k=1}^{n} b_k^2 \sum_{k=1}^{n} c_k^2.$$

1.4. Some Help From Symmetry

a. This pretty much follows from Problem 1.1. Consider,

$$\left(\frac{x+y}{x+y+z}\right)^{\frac{1}{2}} + \left(\frac{x+z}{x+y+z}\right)^{\frac{1}{2}} + \left(\frac{y+z}{x+y+z}\right)^{\frac{1}{2}} \leq 3^{\frac{1}{2}}\left(\frac{2(x+y+z)}{x+y+z}\right)^{\frac{1}{2}} = 6^{\frac{1}{2}}.$$

a. Again, we use the Cauchy-Schwarz inequality,

$$\begin{aligned}
(x+y+z)^2 &= \left(\frac{x}{\sqrt{y+z}}\sqrt{y+z} + \frac{y}{\sqrt{x+z}}\sqrt{x+z} + \frac{z}{\sqrt{x+y}}(\sqrt{x+y})\right)^2 \\
&\leq 2(x+y+z)\left(\frac{x^2}{y+z} + \frac{y^2}{x+z} + \frac{z^2}{x+y}\right) \\
\implies x+y+z &\leq 2\left(\frac{x^2}{y+z} + \frac{y^2}{x+z} + \frac{z^2}{x+y}\right).
\end{aligned}$$

1.5. A Crystallographic Inequality with a Message

Consider, using the Cauchy-Schwarz inequality,

$$\begin{aligned}
g^2(x) &= \left(\sum_{k=1}^{n} p_k^{\frac{1}{2}} p_k^{\frac{1}{2}} \cos(\beta_k x)\right) \\
&\leq \sum_{k=1}^{n} p_k \sum_{k=1}^{n} p_k \cos^2(\beta_k x) \\
&= \sum_{k=1} \frac{p_k}{2}(1+\cos(2\beta_k x)) \\
&\leq \frac{1}{2}\left(\sum_{k=1}^{n} p_k + \sum_{k=1}^{n} p_k \cos(2\beta_k x)\right) \\
&= \frac{1}{2}(1+g(2x)).
\end{aligned}$$

1.6. A Sum of Inversion Preserving Summands

Momentarily, we shift focus on bounding $\sum_{k=1}^{n} \frac{1}{p_k}$ from below. This is also a consequence of Cauchy-Schwarz,

$$\sum_{k=1}^{n} \frac{1}{p_k} = \sum_{k=1}^{n} p_k \sum_{k=1}^{n} \frac{1}{p_k} \geq \left(\sum_{k=1}^{n} 1\right)^2 = n^2 \tag{1}$$

The equality occurs iff $\sqrt{p_k} = \frac{\alpha}{\sqrt{p_k}}$. That is, when all p_k are equal. This then implies that equality occurs iff $p_k = \frac{1}{n}$.

Then, by the Cauchy-Schwarz inequality and the above result,

$$\begin{aligned}
\sum_{k=1}^{n} \left(p_k + \frac{1}{p_k}\right)^2 &\geq \frac{1}{n}\left(\sum_{k=1}^{n} p_k + \sum_{k=1}^{n} \frac{1}{p_k}\right)^2 = \frac{1}{n}\left(1 + \sum_{k=1}^{n} \frac{1}{p_k}\right)^2 \\
&\geq \frac{1}{n}(1+n^2)^2 \geq n^3 + 2n + \frac{1}{n}.
\end{aligned} \tag{2}$$

It is easy to check that $p_k = \frac{1}{n}$ for all k is a sufficient condition for equality in (2). For it to be necessary, we just note that $p_k = \frac{1}{n}$ is a necessary condtion for equality in the second application of Cauchy-Schwarz in (2).

1.7. Flexibility of Form

Consider the inner product on $\mathbb{R}^2$ defined by

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 5x_1y_1 + x_1y_2 + x_2y_1 + 3x_2y_2$$

It is an inner product since,

a. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle > 0$ for non-zero $\boldsymbol{x} \in \mathbb{R}^2$

   Consider, if one of the coordinates of $\boldsymbol{x}$ is nonzero then,

$$5x_1^2 + 2x_1x_2 + 3x_2^2 = 4x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0.$$

b. $\langle \alpha\boldsymbol{x} + \boldsymbol{v}, \boldsymbol{y} \rangle = \alpha\langle \boldsymbol{x}, \boldsymbol{y} \rangle + \langle \boldsymbol{v}, \boldsymbol{y} \rangle$ for all $\alpha \in \mathbb{R}$ and $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{v} \in \mathbb{R}^2$

   Consider the following,

$$\begin{aligned}\langle \alpha\boldsymbol{x} + \boldsymbol{v}, \boldsymbol{y} \rangle &= 5(\alpha x_1 + v_1)y_1 + (\alpha x_1 + v_1)y_2 + (\alpha x_2 + v_2)y_1 + 3(\alpha x_2 + v_2)y_2 \\ &= \alpha(5x_1y_1 + x_1y_2 + x_2y_1 + 3x_2y_2) + (5v_1y_1 + v_1y_2 + v_2y_1 + 3v_2y_2) \\ &= \alpha\langle \boldsymbol{x}, \boldsymbol{y} \rangle + \langle \boldsymbol{v}, \boldsymbol{y} \rangle.\end{aligned}$$

c. $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^2$

   This follows from a routine computation,

$$\begin{aligned}\langle \boldsymbol{x}, \boldsymbol{y} \rangle &= 5x_1y_1 + x_1y_2 + x_2y_1 + 3x_2y_2 \\ &= 5y_1x_1 + y_2x_1 + y_1x_2 + 3y_2x_2 \\ &= \langle \boldsymbol{y}, \boldsymbol{x} \rangle.\end{aligned}$$

Then by Cauchy-Schwarz on $\boldsymbol{x} = (x, y)$ and $\boldsymbol{\alpha} = (\alpha, \beta)$, we have

$$5x\alpha + x\beta + y\alpha + 3y\beta \le \left(5x^2 + 2xy + 3y^2\right)^{\frac{1}{2}}\left(5\alpha^2 + 2\alpha\beta + 3\beta^2\right)^{\frac{1}{2}}.$$

1.8. Doing Sums

a. $\sum_{k=0}^{\infty} a_k x^k \le \frac{1}{\sqrt{1-x^2}}\left(\sum_{k=0}^{\infty} a_k^2\right)^{\frac{1}{2}}$ for $x \in [0, 1)$

   For $x \in [0, 1)$, we have $\sum_{k=0}^{\infty} x^{2k} = \frac{1}{1-x^2}$. If $\sum_{k=0}^{\infty} a_k^2 = \infty$ the bound trivially holds. Otherwise, we apply Cauchy-Schwarz to obtain

$$\sum_{k=0}^{\infty} a_k x^k \le \frac{1}{\sqrt{1-x^2}}\left(\sum_{k=0}^{\infty} a_k^2\right)^{\frac{1}{2}}.$$

b. $\sum_{k=1}^{n} \frac{a_k}{k} < \sqrt{2}\left(\sum_{k=1}^{n} a_k^2\right)^{\frac{1}{2}}$

   We start by noting,

$$\sum_{k=1}^{n} \frac{1}{k^2} \le 1 + \int_1^n \frac{1}{k^2}dk = 2 - \frac{1}{n} < 2.$$

   When $\sum_{k=0}^{\infty} a_k^2 = \infty$ the bound trivially holds. Otherwise, we apply Cauchy-Schwarz to obtain

$$\sum_{k=0}^{\infty} \frac{a_k}{k} < \sqrt{2}\left(\sum_{k=1}^{n} a_k^2\right)^{\frac{1}{2}}.$$

c. $\sum_{k=1}^{n} \frac{a_k}{\sqrt{n+k}} < (\log 2)^{\frac{1}{2}} \left(\sum_{k=1}^{n} a_k^2\right)^{\frac{1}{2}}$

We start by noting that

$$\sum_{k=1}^{n} \frac{1}{n+k} < \int_{n}^{2n} \frac{1}{k} dk = \ln 2n - \ln n = \ln 2.$$

When $\sum_{k=0}^{\infty} a_k^2 = \infty$ the bound trivially holds. Otherwise, we apply Cauchy-Schwarz to obtain

$$\sum_{k=1}^{n} \frac{a_k}{\sqrt{n+k}} < (\ln 2)^{\frac{1}{2}} \left(\sum_{k=1}^{n} a_k^2\right)^{\frac{1}{2}}.$$

d. $\sum_{k=0}^{n} \binom{n}{k} a_k \leq \binom{2n}{n}^{\frac{1}{2}} \left(\sum_{k=0}^{n} a_k^2\right)^{\frac{1}{2}}$

This is a standard application of Cauchy-Schwarz after proving that $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$. This is equivalent to $\sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$. We give a combinatorial proof of this. The right hand side counts the number of ways of choosing n elements of a $2n$-element set. To see that the left hand side also does so, consider the following process of selection: fix a partition of a $2n$-element set into two sets of size n; from the first partition select k elements and from the second parition select $n-k$ elements. Summing up over all choices of k, we get the left hand side.
Finally, apply Cauchy-Schwarz to obtain

$$\sum_{k=0}^{n} \binom{n}{k} a_k \leq \left(\sum_{k=0}^{n} \binom{n}{k}^2\right)^{\frac{1}{2}} \left(\sum_{k=0}^{n} a_k^2\right)^{\frac{1}{2}} = \binom{2n}{n}^{\frac{1}{2}} \left(\sum_{k=0}^{n} a_k^2\right)^{\frac{1}{2}}.$$

1.9. Beating the Obvious Bounds

First, note that $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$. Thus, we get

$$\begin{aligned}
\left|\sum_{j=1}^{n} a_j\right|^2 + \left|\sum_{j=1}^{n} (-1)^j a_j\right|^2 &= 2\left(\left(\sum_{j=1}^{\lfloor n/2 \rfloor} a_{2j}\right)^2 + \left(\sum_{j=1}^{\lceil \frac{n}{2} \rceil} a_j\right)^2\right) \\
&\leq 2\left(\left\lfloor \frac{n}{2} \right\rfloor \sum_{j=1}^{\lfloor n/2 \rfloor} a_{2j}^2 + \left\lceil \frac{n}{2} \right\rceil \sum_{j=1}^{\lceil n/2 \rceil} a_j^2\right) \\
&\leq 2\left(\left(\frac{n}{2}+1\right) \sum_{j=1}^{\lceil n/2 \rceil} a_{2j}^2 + \left(\frac{n}{2}+1\right) \sum_{j=1}^{\lceil n/2 \rceil} a_j^2\right) \\
&\leq (n+2) \sum_{j=1}^{n} a_j^2.
\end{aligned}$$

1.10. Schur's Lemma – The R and C Bound

$$
\begin{aligned}
\left|\sum_{j=1}^{m}\sum_{k=1}^{n} c_{jk}x_j y_k\right|^2 &\leq \left(\sum_{j=1}^{m} x_j^2\right)\left(\sum_{j=1}^{m}\left(\sum_{k=1}^{n} c_{j,k}y_k\right)^2\right) \\
&\leq \left(\sum_{j=1}^{m} x_j^2\right)\left(\sum_{j=1}^{m}\left(\sum_{k=1}^{n} c_{jk}\right)\left(\sum_{k=1}^{n} c_{jk}y_k^2\right)\right) \\
&\leq \left(\max_k \sum_{k=1}^{n}|c_{jk}|\right)\left(\sum_{j=1}^{m} x_j^2\right)\left(\sum_{k=1}^{n}\sum_{j=1}^{m} c_{jk}y_k^2\right) \\
&\leq \left(\max_k \sum_{k=1}^{n}|c_{jk}|\right)\left(\max_j \sum_{j=1}^{m} c_{jk}\right)\left(\sum_{j=1}^{m} x_j^2\right)\left(\sum_{k=1}^{n} y_k^2\right) \\
&= CR\left(\sum_{j=1}^{m} x_j^2\right)\left(\sum_{k=1}^{n} y_k^2\right).
\end{aligned}
$$

1.11. Schwarz's Argument in an Inner Product Space

$$p(t) = \langle \boldsymbol{v} + t\boldsymbol{w}, \boldsymbol{v} + t\boldsymbol{w}\rangle = \|\boldsymbol{v}\|^2 + 2t\langle \boldsymbol{v}, \boldsymbol{w}\rangle + t^2\|\boldsymbol{w}\|^2$$

We know that $p(t) \geq 0$. Thus, the discriminant must be non-positive. That is,

$$(2\langle \boldsymbol{v}, \boldsymbol{w}\rangle)^2 - 4\|\boldsymbol{v}\|^2\|\boldsymbol{w}\|^2 \leq 0 \Longrightarrow \langle \boldsymbol{v}, \boldsymbol{w}\rangle \leq \|\boldsymbol{v}\|\|\boldsymbol{w}\|.$$

Equality holds if and only if the discriminant is 0. This is true if and only if there exists a $t^\star$ such that $p(t^\star) = 0$. In turn, this holds if and only if $\|v + t^\star\boldsymbol{w}\| = 0$ which is true if and only if $\boldsymbol{v} = -t^\star\boldsymbol{w}$.

1.12. Example of a Self-Generalization

Consider the vector space V^n, with the inner product

$$\langle (\boldsymbol{x}_1, ..., \boldsymbol{x}_n), (\boldsymbol{y}_1, ..., \boldsymbol{y}_n)\rangle = \sum_{i=1}^{n}\langle \boldsymbol{x_i}, \boldsymbol{y_i}\rangle.$$

It is easy to verify that this is in fact an inner product. Then, by the fact that Cauchy-Schwarz holds in abstract inner product spaces, we then have:

$$
\begin{aligned}
\sum_{i=1}^{n}\langle \boldsymbol{x}_i, \boldsymbol{y}_i\rangle &\leq \langle (\boldsymbol{x}_1, ..., \boldsymbol{x}_n), (\boldsymbol{y}_1, ..., \boldsymbol{y}_n)\rangle \\
&\leq \langle (\boldsymbol{x}_1, ..., \boldsymbol{x}_n), (\boldsymbol{x}_1, ..., \boldsymbol{x}_n)\rangle^{\frac{1}{2}}\langle (\boldsymbol{y}_1, ..., \boldsymbol{y}_n), (\boldsymbol{y}_1, ..., \boldsymbol{y}_n)\rangle^{\frac{1}{2}} \\
&\leq \left(\sum_{i=1}^{n}\langle \boldsymbol{x}_i, \boldsymbol{x}_i\rangle\right)^{\frac{1}{2}}\left(\sum_{i=1}^{n}\langle \boldsymbol{y}_i, \boldsymbol{y}_i\rangle\right)^{\frac{1}{2}}.
\end{aligned}
$$

1.13. Application of Cauchy's Inequality to an Array

Let $b_{jk} = \alpha_j + \beta_k$ with

$$\alpha_j = \frac{1}{n}\sum_{k=1}^{n} a_{jk} \qquad \beta_k = \frac{1}{m}\sum_{j=1}^{m} a_{jk}.$$

Without loss of generality we may suppose $\sum_{j,k} a_{jk} = \sum_{[m]} \alpha_j = \sum_{[n]} \beta_k = 0$. Then note,

$$
\begin{aligned}
0 &\le \sum_{j,k} \left(a_{jk} - b_{jk}\right)^2 \\
&= \sum_{j,k} \left(a_{jk}^2 - 2a_{jk}b_{jk} + b_{jk}^2\right) \\
&= \sum_{j,k} a_{jk}^2 - 2\sum_{j,k} a_{jk}\alpha_j - 2\sum_{j,k} a_{jk}\beta_k + \sum_{j,k} \left(\alpha_j + \beta_k\right)^2 \\
&= \sum_{j,k} a_{jk}^2 - 2n\sum_j \alpha_j^2 - 2m\sum_k \beta_k^2 + n\sum_j \alpha_j^2 + 2\sum_{j,k} \alpha_j\beta_k + m\sum_k \beta_k^2 \\
\implies \frac{1}{n}\sum_j \alpha_j^2 + \frac{1}{m}\sum_k \beta_k^2 &\le \sum_{j,k} a_{jk}^2.
\end{aligned}
$$

The general result follows after an appropriate change of variables.

1.14. A Cauchy Triple and Loomis-Whitney

a. Consider,

$$
\begin{aligned}
\sum_{i,j,k=1}^{n} a_{ij}^{\frac{1}{2}} b_{jk}^{\frac{1}{2}} c_{ki}^{\frac{1}{2}} &= \sum_{i,j=1}^{n} a_{ij}^{\frac{1}{2}} \sum_{k=1}^{n} b_{jk}^{\frac{1}{2}} c_{ki}^{\frac{1}{2}} \\
&\le \sum_{i,j=1}^{n} a_{ij}^{\frac{1}{2}} \left(\sum_{k=1}^{n} b_{jk}\right)^{\frac{1}{2}} \left(\sum_{k=1}^{n} c_{ki}\right)^{\frac{1}{2}} \\
&= \sum_{j=1}^{n} \left(\sum_{k=1}^{n} b_{jk}\right)^{\frac{1}{2}} \sum_{i=1}^{n} a_{ij}^{\frac{1}{2}} \left(\sum_{k=1}^{n} c_{ki}\right)^{\frac{1}{2}} \\
&\le \sum_{j=1}^{n} \left(\sum_{k=1}^{n} b_{jk}\right)^{\frac{1}{2}} \left(\sum_{i=1}^{n} a_{ij}\right)^{\frac{1}{2}} \left(\sum_{i,k=1}^{n} c_{ki}\right)^{\frac{1}{2}} \\
&\le \left(\sum_{i,j=1}^{n} a_{ij}\right)^{\frac{1}{2}} \left(\sum_{j,k=1}^{n} b_{jk}\right)^{\frac{1}{2}} \left(\sum_{i,k=1}^{n} c_{ki}\right)^{\frac{1}{2}}.
\end{aligned}
$$

Consider,
$\sum_{i,j,k=1}^{n}$

b. Let $n = \max_{\boldsymbol{x} \in A} \|\boldsymbol{x}\|_\infty$. Then let

$$
\begin{aligned}
a_{ij} &= \mathbb{1}\{(0, i, j) \in A_x\}, \\
b_{jk} &= \mathbb{1}\{(k, 0, j) \in A_y\}, \\
c_{ki} &= \mathbb{1}\{(k, i, 0) \in A_z\}.
\end{aligned}
$$

Note that $a_{ij}^{\frac{1}{2}} b_{jk}^{\frac{1}{2}} c_{ki}^{\frac{1}{2}} = a_{ij} b_{jk} c_{ki} \in \{0, 1\}$. Furthermore, $a_{ij} b_{jk} c_{ki} = 1$ if and only if $(i, j, k) \in A$. Thus,

$$
|A| = \sum_{i,j,k=1}^{n} a_{ij}^{\frac{1}{2}} b_{jk}^{\frac{1}{2}} c_{ki}^{\frac{1}{2}} \le \left(\sum_{i,j=1}^{n} a_{ij}\right)^{\frac{1}{2}} \left(\sum_{j,k=1}^{n} b_{jk}\right)^{\frac{1}{2}} \left(\sum_{i,k=1}^{n} c_{ki}\right)^{\frac{1}{2}} = |A_x|^{\frac{1}{2}}\ |A_y|^{\frac{1}{2}}\ |A_z|^{\frac{1}{2}}.
$$

1.15. An Application to Statistical Theory

First, we differentiate with respect to θ.

$$\sum_{k\in D} g(k)p(k;\theta) = \theta \Rightarrow \sum_{k\in D} g(k)p_\theta(k;\theta) = 1$$
$$\sum_{k\in D} p(k;\theta) = 1 \Rightarrow \sum_{k\in D} p_\theta(k;\theta) = 0$$

Now, we apply Cauchy-Schwarz,

$$\begin{aligned}
1 &= \left(\sum_{k\in D}(g(k)-\theta)p_\theta(k;\theta)\right)^2 \\
&\le \left(\sum_{k\in D}(g(k)-\theta)^2 p(k;\theta)\right)\left(\sum_{k\in D}\left(\frac{p_\theta(k;\theta)}{p(k;\theta)}\right)^2 p_\theta(k;\theta)\right) \\
&\le \mathbf{Var}[g]\cdot \mathbf{I}(\theta).
\end{aligned}$$

2. The AM-GM Bound

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